



**Faculty of Engineering and Technology Department of Electrical and  
Computer Engineering**

**Software Engineering – COMP433**

(Second Semester – Summer 2025-2026)

**Vet Doctor**

**Final Report**

**Group: 10**

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**Prof: Dr. Adel Taweel**

Date: June 25, 2026

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## Group Members



Noor Nasir



Wafaa Shaheen



Ellen Izz



Fadi Bassous



Yousef Qawwas

# **1. Project Planning and Management**

## **1.1 Writers of the Report**

The final project report was edited and organized by:

- Noor Nasir
- Yousef Qawwas
- Wafaa Shaheen

## **1.2 Business Title**

Vet Doctor

## **1.3 Group Name**

G10

## **1.4 Group Members**

| <b>Name</b>   | <b>University ID</b> |
|---------------|----------------------|
| Noor Nasir    | 1220769              |
| Yousef Qawwas | 1220137              |
| Ellen Izz     | 1220806              |
| Fadi Bassous  | 1221005              |
| Wafaa Shaheen | 1221769              |

## **1.5 Role of Each Member**

- Noor Nasir (1220769) – Project Manager
- Yousef Qawwas (1220137) – Technical Architect
- Ellen Izz (1220806) – Requirement Engineer
- Fadi Bassous (1221005) – Programmer
- Wafaa Shaheen (1221769) – Q/A / Software Tester

## **1.6 Project Management Strategy**

Managing a five-member team across multiple project phases required a structured yet flexible approach. The backbone of the team's coordination was consistent communication through messaging applications, which allowed members to stay informed, raise concerns, and share updates without needing to schedule a formal meeting for every small decision. This kept momentum going between phases and reduced delays caused by miscommunication.

Formal meetings were scheduled at critical points in the project timeline, primarily when a new phase was beginning, or a submission deadline was approaching. The Project Manager took responsibility for tracking each member's academic commitments and availability and used this awareness to distribute tasks in a way that was realistic and fair rather than simply equal on paper. When a member was under pressure from other coursework, this was taken into account during task assignment. The goal was always to maintain quality without placing an unreasonable burden on any individual.

Task ownership was clearly defined from the start of each phase, with each member contributing according to their designated role. The Project Manager reviewed all deliverables before submission to ensure consistency and coherence across individual contributions.

When decisions needed to be made, the team discussed options openly and aimed for agreement. If no clear consensus emerged, the Project Manager made the final call to keep the project moving forward without unnecessary delays.

Regarding the software process model, the project naturally followed a hybrid approach. The overall structure was sequential, each phase had defined inputs and outputs that fed into the next reflecting a Waterfall model at the macro level. Within individual phases however, the team worked iteratively, incorporating instructor feedback and refining outputs before carrying them forward, which reflects Agile thinking at the micro level.

For UML modelling, the team used Enterprise Architect, which provided the notation support and structure needed to produce the diagrams required across the project phases.

## **1.7 Project Manager Report**

### **Contributions by Each Group Member**

Ellen Izz, as Requirements Engineer, led the user requirements analysis and authored the Book Appointment use case specification. Yousef Qawwas, as Technical Architect, led the system requirements and took ownership of the Make Payment use case and the activity diagram. Wafaa Shaheen, as QA and Software Tester, led the actor analysis and provided reviewing support consistently throughout the project. Fadi Bassous, as Programmer, led the use case modelling task and authored the Submit Review use case specification. As Project Manager, I coordinated task distribution, led the effort and time estimation, authored the Log In use case specification, and reviewed all deliverables before submission.

### **Barriers and Challenges**

The most persistent challenge was coordinating five members with different schedules and workloads across a full semester. Reaching consensus on meeting times was often difficult, and progress frequently relied on asynchronous communication. A recurring challenge was consistently meeting the instructor's precise standards for formal requirements writing — producing unambiguous, verifiable, and correctly structured documentation proved demanding for the entire team. Producing correct UML diagrams was also more technically challenging than anticipated, with notation and relationship errors directly impacting submission quality.

### **Evaluation of Project Success**

The team showed clear improvement from Phase 1 through to the current phase, with each submission reflecting a better understanding of software engineering principles. Every deadline was met and instructor feedback was taken seriously at each stage. The main lesson learned was that investing time early in understanding formal documentation standards would have significantly improved the quality of earlier phases. Overall, I consider this project a valuable and honest learning experience, and I am confident the final report will reflect the full extent of what the team has developed throughout this semester.

## **1.8 Group Members Reports**

### **Noor Nasir**

Throughout this project, I took on the role of Project Manager, which meant I was responsible for coordinating the team, distributing tasks, tracking progress, and ensuring that every phase was submitted on time and in a coherent form. Beyond the management responsibilities, I contributed directly to the technical work by leading the effort and time estimation task, authoring the Log In use case specification, and drawing the activity diagram in Phase 3. I also reviewed all group deliverables before submission to ensure consistency across individual contributions. Evaluating the project overall, I believe the team made genuine progress with each phase, and the feedback we received, while challenging at times, pushed us to develop a more rigorous and precise approach to software engineering documentation. The experience of managing a group through a multi-phase project taught me a great deal about communication, fairness, and keeping a team aligned under pressure, and I am satisfied with how the team performed given the challenges we faced.

### **Wafaa Shaheen**

Throughout this project, I worked as the Q/A and Software Tester, focusing on reviewing the project deliverables, checking consistency between requirements and diagrams, and helping ensure that the final documentation followed the instructor's feedback and required format. In Phase 2, I led the Actors Analysis task and contributed to the User Requirements, System Requirements, and Effort and Time Estimation by reviewing, discussing, and finalizing the written sections. In Phase 3, I authored the Acknowledge Emergency Booking use case specification and contributed to reviewing the use-case model and activity diagram. In the System Analysis and Modelling phase, I contributed to the class diagrams and worked on the Acknowledge Emergency Booking sequence diagram, making sure it matched the use case description and the classes in the detailed class model. Overall, I believe the project was successful because our team improved the quality of the work across phases by applying feedback, correcting modelling issues, and making the final report more coherent. My main contribution was ensuring that the system behavior, requirements, and diagrams were clear, consistent, and properly reviewed before submission.

### **Fadi Bassous**

I contributed to the system analysis and modelling part of the Vet Doctor project. I led the work on the System Class Diagrams section, including both the Analysis Class Model and the Detailed Class Model. My work focused on identifying the main domain classes, defining the relationships between them, adding multiplicities, and refining the detailed class diagram by adding attributes, operations, datatypes, and enumerations. I also worked on my individual

Sequence Diagram task, where I used the approved class diagrams and the use-case flow to model the interaction between the system objects clearly and consistently with UML notation.

### **Ellen Izz**

Throughout this project, I worked as the Requirement Engineer, focusing on understanding the Vet Doctor business needs and translating them into clear and structured software requirements. In Phase 2, I led the User Requirements task and contributed to reviewing and discussing the System Requirements, Effort and Time Estimation, and Actors Analysis sections. I also authored the Book Appointment use case specification, where I described how a client requests a veterinary home or farm visit, how the system checks availability, assigns a veterinarian, creates the appointment, and sends the required notifications. In Phase 3, I contributed to the Use-Case Diagram and Activity Diagram by discussing and reviewing the flow of the system to ensure it matched the written requirements. In the System Analysis and Modelling phase, I contributed to the class diagrams and worked on the Book Appointment sequence diagram. Overall, I believe our project was successful because the team improved its work across the different phases and used the feedback received to make the final report more complete and consistent. My main contribution was to ensure that the system requirements and use-case descriptions clearly represented the client's needs and supported the later diagrams and design decisions.

### **Yousef Qawwas**

Throughout this project, I worked as a Technical Architect, focusing on keeping the requirements, models, and design decisions consistent across all phases. In Phase 2, I led the System Requirements task and participated in the User Requirements and Effort and Time Estimation discussions. In Phase 3, I contributed to Actor Analysis and Use-Case Modelling, wrote the detailed Make Payment use case, and led the Activity Diagram task. In Phase 4, I worked on the Make Payment sequence diagram and contributed to the architectural design based on the system design goals. Finally, I contributed to writing this final report. Overall, I believe our team successfully turned the Vet Doctor business idea into a clear software engineering model by iterating on the feedback we received. My contributions focused on maintaining consistency across each phase, ensuring a seamless connection between the project requirements, behavioral diagrams, and system architecture.



## 2. Requirement Elicitation, Analysis and Modeling

### 2.1 Requirement Statement and Business Description

The system developed for this project is commissioned by **Vet Doctor**, a veterinary home visit service that connects pet owners and farmers with licensed veterinary doctors who travel directly to the customer's home or farm to treat their animals. The business previously managed its bookings manually through phone calls and basic tools, which limited its operational efficiency, restricted bookings to working hours only, and provided no real-time tracking of appointments or veterinarian activity. The client requested the development of a fully integrated online platform to address these limitations and improve service management, accessibility, and overall customer experience.

The business serves two main categories of customers: pet owners who keep animals such as dogs, cats, birds, and rabbits, and farmers who raise livestock including sheep, goats, cows, and chickens. The service model is built around sending the veterinarian to the customer rather than requiring the customer to travel to a clinic, which is the defining characteristic of the business.

The services the system supports include home visit booking, emergency visit requests, routine check-ups, online payment processing, veterinarian profile browsing, and online consultation. The business operates with a team consisting of an administrator, a number of veterinary doctors, and customer support staff, with the system designed to scale as the number of users and veterinarians grows over time.

Revenue is generated through fixed service pricing per visit type, optional packages or subscription plans, and additional charges for medications, laboratory tests, or specialized treatments. The business collects payments from customers and distributes earnings to veterinarians while retaining a profit margin.

The system unifies booking management, veterinarian coordination, payment processing, customer feedback, and performance reporting into a single platform, replacing the previous manual processes with an efficient, accessible, and scalable solution.

## 2.2 User and System Requirement

MODIFIED

### User Requirements

Lead: Ellen Izz

Contributors: [Noor Nasir – reviewing], [Yousef Qawwas – discussion], [Wafaa Shaheen – finalising]

- **UR1.0:** The Vet Doctor system shall enable clients and veterinarians to register accounts.
- **UR2.0:** Registered users shall be able to log in to the Vet Doctor system using their registered credentials.
- **UR3.0:** The Vet Doctor system shall enable clients to book veterinary home or farm visits.
- **UR4.0:** Registered clients shall be able to browse veterinarian profiles.
- **UR5.0:** Registered veterinarians shall be able to manage their assigned veterinary visits.
- **UR6.0:** Registered clients shall be able to make payments for veterinary services.
- **UR7.0:** Registered clients shall be able to follow up their booked appointments with the assigned veterinarian.
- **UR8.0:** The administrator shall be able to oversee Vet Doctor business operations.
- **UR9.0:** Registered clients shall be able to submit feedback for completed veterinary visits.
- **UR10.0:** Registered clients shall be able to update their account information.

## System Requirements

**Lead:** Yousef Qawwas

**Contributors:** [Ellen Izz – reviewing], [Fadi Bassous – discussion], [Wafaa Shaheen – finalising]

**UR1.0:** The Vet Doctor system shall enable clients and veterinarians to register accounts.

- **SR1.1:** The system shall enable a client to create an account by providing full name, phone number, email address, physical address, client type, and password.
- **SR1.2:** The system shall enable a veterinarian to create an account by providing full name, phone number, email address, password, professional license number, specialization, and service area.
- **SR1.3:** The system shall validate the email address format before creating a new account.
- **SR1.4:** The system shall require each registered account to have a unique email address.
- **SR1.5:** The system shall reject a registration request when any required registration field is missing.
- **SR1.6:** The system shall require account passwords to contain at least 8 characters, including at least one uppercase letter, one digit, and one special character.
- **SR1.7:** The system shall store account passwords in a non-reversible encrypted form.
- **SR1.8:** The system shall restrict administrator account creation to the system owner.

**UR2.0:** Registered users shall be able to log in to the Vet Doctor system using their registered credentials.

- **SR2.1:** The system shall enable registered users to submit their registered email address and password for login.
- **SR2.2:** The system shall authenticate a login request by verifying the submitted email address and password.
- **SR2.3:** The system shall reject a login request when the submitted email address or password is incorrect.
- **SR2.4:** The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts.

- **SR2.5:** The system shall notify the user by email when the account is locked after failed login attempts.
- **SR2.6:** The system shall enable registered users to request a password reset link through their registered email address.
- **SR2.7:** The system shall make each password reset link valid for 30 minutes.
- **SR2.8:** The system shall require a new password reset request after the password reset link expires.

**UR3.0:** The Vet Doctor system shall enable clients to book veterinary home or farm visits.

- **SR3.1:** The system shall enable a logged-in client to create a booking request for Routine Check-Up, Emergency Visit, or Farm Visit.
- **SR3.2:** The system shall require each booking request to include the animal type.
- **SR3.3:** The system shall require each booking request to include a short description of the visit reason.
- **SR3.4:** The system shall require each booking request to include the visit location.
- **SR3.5:** The system shall require each booking request to include a preferred appointment date and time.
- **SR3.6:** The system shall accept appointment bookings only for available time slots.
- **SR3.7:** The system shall enable the client to choose a veterinarian whose specialization and service area match the booking request.
- **SR3.8:** The system shall assign a matching available veterinarian when the client does not choose a specific veterinarian.
- **SR3.9:** The system shall classify Emergency Visit bookings as high-priority bookings.
- **SR3.10:** The system shall notify the assigned veterinarian of a new Emergency Visit booking within 1 minute of assignment.

- **SR3.11:** The system shall mark an Emergency Visit booking as unacknowledged if the assigned veterinarian does not respond within 15 minutes.
- **SR3.12:** The system shall enable the client to cancel a booked appointment before its scheduled time.
- **SR3.13:** The system shall enable the client to reschedule a booked appointment before its scheduled time.
- **SR3.14:** The system shall notify the client and the assigned veterinarian when an appointment is cancelled or rescheduled.
- **SR3.15:** The system shall notify the client of the cancellation fee when the appointment is cancelled less than 2 hours before the scheduled visit time.

**UR4.0:** Registered clients shall be able to browse veterinarian profiles.

- **SR4.1:** The system shall provide registered clients with veterinarian profile information including full name, years of experience, specialization, and service area.
- **SR4.2:** The system shall provide registered clients with each veterinarian's availability status for the current day.
- **SR4.3:** The system shall provide registered clients with each veterinarian's available appointment slots for the next 30 days.
- **SR4.4:** The system shall mark already booked veterinarian time slots as unavailable.
- **SR4.5:** The system shall provide registered clients with each veterinarian's average rating.
- **SR4.6:** The system shall provide registered clients with the most recent 5 verified reviews for each veterinarian.
- **SR4.7:** The system shall enable registered clients to search veterinarians by name or specialization.
- **SR4.8:** The system shall enable registered clients to filter veterinarians by specialization, available date, minimum rating, and service location.
- **SR4.9:** The system shall notify the client when no veterinarians match the selected search or filter criteria.

**UR5.0:** Registered veterinarians shall be able to manage their assigned veterinary visits.

- **SR5.1:** The system shall provide each logged-in veterinarian with their assigned appointments sorted by scheduled date and time.
- **SR5.2:** The system shall provide the veterinarian with the service type, animal type, client name, visit location, and scheduled time for each assigned appointment.
- **SR5.3:** The system shall enable the veterinarian to set available and unavailable time slots.
- **SR5.4:** The system shall prevent new bookings for veterinarian time slots marked as unavailable.
- **SR5.5:** The system shall enable the veterinarian to update the appointment status using the stages Confirmed, En Route, In Progress, and Completed.
- **SR5.6:** The system shall notify the client within 30 seconds after the assigned veterinarian updates the appointment status.
- **SR5.7:** The system shall enable the veterinarian to submit post-visit notes after marking a visit as Completed.
- **SR5.8:** The system shall limit post-visit notes to 2,000 characters.
- **SR5.9:** The system shall enable the veterinarian to add a prescription record for a completed visit.
- **SR5.10:** The system shall enable the veterinarian to attach files related to a completed visit.
- **SR5.11:** The system shall reject attached visit files larger than 10 MB per file.
- **SR5.12:** The system shall associate post-visit records with the related animal record.
- **SR5.13:** The system shall restrict veterinarian access to animal records linked to their assigned appointments.
- **SR5.14:** The system shall enable the veterinarian to set a follow-up date for a completed appointment.
- **SR5.15:** The system shall send a follow-up reminder to the client and veterinarian

on the specified follow-up date.

**UR6.0:** Registered clients shall be able to make payments for veterinary services.

- **SR6.1:** The system shall support cash payment for veterinary services.
- **SR6.2:** The system shall support online payment via Credit/Debit Card service for veterinary services.
- **SR6.3:** The system shall record the selected payment method for each booking.
- **SR6.4:** The system shall calculate the initial service charge based on the selected service type.
- **SR6.5:** The system shall enable the veterinarian to add post-visit charges for medications, laboratory tests, or extended service time.
- **SR6.6:** The system shall require client acknowledgment before adding post-visit charges to the final invoice.
- **SR6.7:** The system shall generate a digital invoice after a successful online payment.
- **SR6.8:** The system shall generate a digital invoice after the veterinarian confirms cash receipt.
- **SR6.9:** The system shall include the service date, veterinarian name, service type, itemized charges, and total amount in each invoice.
- **SR6.10:** The system shall send the generated invoice to the client's registered email address.
- **SR6.11:** The system shall keep a payment history for each client.
- **SR6.12:** The system shall keep a payment history for each veterinarian.
- **SR6.13:** The system shall enable users to filter their payment history by date range.
- **SR6.14:** The system shall enable users to download invoices as PDF files.
- **SR6.15:** The system shall not store full card numbers for online payments.

- **SR6.16:** The system shall store only a masked card reference for online payment records.

**UR7.0:** Registered clients shall be able to follow up their booked appointments with the assigned veterinarian.

- **SR7.1:** The system shall send a booking confirmation to the client and assigned veterinarian within 30 seconds after appointment assignment.
- **SR7.2:** The system shall include appointment date, time, location, and service type in the booking confirmation.
- **SR7.3:** The system shall provide the client with the current appointment status.
- **SR7.4:** The system shall make appointment status updates available to the client within 30 seconds after the veterinarian changes the status.
- **SR7.5:** The system shall send a reminder to the client 24 hours before the scheduled appointment.
- **SR7.6:** The system shall send a reminder to the client 1 hour before the scheduled appointment.
- **SR7.7:** The system shall provide a messaging channel between the client and the assigned veterinarian for confirmed or active appointments.
- **SR7.8:** The system shall restrict appointment messaging to the client and veterinarian assigned to that appointment.
- **SR7.9:** The system shall support text messages between the client and the assigned veterinarian.
- **SR7.10:** The system shall support image attachments in appointment messages.
- **SR7.11:** The system shall retain appointment communication history for 12 months.
- **SR7.12:** The system shall enable the client to request an online consultation for an active appointment.
- **SR7.13:** The system shall enable the assigned veterinarian to accept or decline an online consultation request.



- **SR7.14:** The system shall start a voice or video consultation session after the assigned veterinarian accepts the consultation request.

**UR8.0:** The administrator shall be able to oversee Vet Doctor business operations.

- **SR8.1:** The system shall provide the administrator with the number of active clients.
- **SR8.2:** The system shall provide the administrator with the number of registered veterinarians.
- **SR8.3:** The system shall provide the administrator with the number of bookings scheduled for the current day.
- **SR8.4:** The system shall provide the administrator with the number of pending payments.
- **SR8.5:** The system shall enable the administrator to review new veterinarian registration requests before activation.
- **SR8.6:** The system shall require administrator approval before activating a veterinarian account.
- **SR8.7:** The system shall enable the administrator to verify veterinarian professional credentials before account activation.
- **SR8.8:** The system shall enable the administrator to change a user account status to active, suspended, deactivated, or deleted.
- **SR8.9:** The system shall record each administrator account-status action with the administrator's name, action type, reason, and timestamp.
- **SR8.10:** The system shall generate a monthly report showing bookings by service type.
- **SR8.11:** The system shall generate a monthly report showing total revenue.
- **SR8.12:** The system shall generate a monthly report showing revenue contribution by veterinarian.
- **SR8.13:** The system shall generate a monthly report showing client acquisition and retention rates.
- **SR8.14:** The system shall enable the administrator to export monthly reports as PDF or spreadsheet files.

- **SR8.15:** The system shall enable the administrator to update service pricing for each service type.
- **SR8.16:** The system shall enable the administrator to update supported payment methods.
- **SR8.17:** The system shall enable the administrator to update notification message templates.

**UR9.0:** Registered clients shall be able to submit feedback for completed veterinary visits.

- **SR9.1:** The system shall request feedback from the client within 24 hours after a visit is marked as Completed.
- **SR9.2:** The system shall expire the feedback request 72 hours after it is sent.
- **SR9.3:** The system shall enable the client to submit one rating from 1 to 5 for each completed appointment.
- **SR9.4:** The system shall enable the client to submit one optional written review for each completed appointment.
- **SR9.5:** The system shall limit each written review to 500 characters.
- **SR9.6:** The system shall prevent feedback submission for cancelled appointments.
- **SR9.7:** The system shall prevent feedback submission for incomplete appointments.
- **SR9.8:** The system shall prevent more than one feedback submission for the same completed appointment.
- **SR9.9:** The system shall associate each feedback submission with its appointment and veterinarian.
- **SR9.10:** The system shall check written reviews for inappropriate language before publication.
- **SR9.11:** The system shall hold flagged reviews for administrator approval before publication.

- **SR9.12:** The system shall make approved reviews available on the related veterinarian profile within 1 hour of approval.
- **SR9.13:** The system shall calculate the veterinarian's average rating after each approved
- **SR9.14:** The system shall enable the administrator to remove a review after recording a removal reason.
- **SR9.15:** The system shall recalculate the veterinarian's average rating after review removal

**UR10.0:** Registered clients shall be able to update their account information.

- **SR10.1:** The system shall enable a logged-in client to update their full name.
- **SR10.2:** The system shall enable a logged-in client to update their phone number.
- **SR10.3:** The system shall enable a logged-in client to update their physical address.
- **SR10.4:** The system shall enable a logged-in client to update their client type.
- **SR10.5:** The system shall enable a logged-in client to request an email address change.
- **SR10.6:** The system shall validate the format of the new email address before saving the change.
- **SR10.7:** The system shall require the new email address to be unique among registered accounts.
- **SR10.8:** The system shall send a verification link to the new email address before replacing the old email address.
- **SR10.9:** The system shall enable a logged-in client to change their password after verifying the current password.
- **SR10.10:** The system shall require the new password to satisfy the password rules defined in SR1.6.
- **SR10.11:** The system shall reject an account update request when any required updated value is invalid.
- **SR10.12:** The system shall notify the client after account information is

successfully updated.

## 2.3 Effort and Time Estimation

MODIFIED

**Lead:** Noor Nasir

**Contributors:** [Wafaa Shaheen – reviewing], [Yousef Qawwas – discussion], [Fadi Bassous – finalising]

| User Requirement                   | Estimated No. of Developers | Estimated Effort Per Developer | Total Effort        |
|------------------------------------|-----------------------------|--------------------------------|---------------------|
| UR1: Account Registration          | 1                           | 2 pw                           | $1 \times 2 = 2$ pw |
| UR2: System Login                  | 1                           | 1 pw                           | $1 \times 1 = 1$ pw |
| UR3: Appointment Booking           | 2                           | 4 pw                           | $2 \times 4 = 8$ pw |
| UR4: Veterinarian Profile Browsing | 1                           | 2 pw                           | $1 \times 2 = 2$ pw |
| UR5: Veterinarian Visit Management | 2                           | 3 pw                           | $2 \times 3 = 6$ pw |

|                                              |                                                                                                 |              |                     |
|----------------------------------------------|-------------------------------------------------------------------------------------------------|--------------|---------------------|
| UR6: Payment Processing                      | 2                                                                                               | 2 pw         | $2 \times 2 = 4$ pw |
| UR7: Appointment Follow-Up and Communication | 2                                                                                               | 3 pw         | $2 \times 3 = 6$ pw |
| UR8: Business Operations Administration      | 2                                                                                               | 2 pw         | $2 \times 2 = 4$ pw |
| UR9: Customer Feedback                       | 1                                                                                               | 2 pw         | $1 \times 2 = 2$ pw |
| UR10: Account Information Update             | 1                                                                                               | 1 pw         | $1 \times 1 = 1$ pw |
| Total Effort / Avg                           | $(1+1+2+1+2+2+2+2+1+1)/10 = 15/10 = 1.5$ devs                                                   | <b>26 pw</b> | <b>34 pw</b>        |
| Schedule Time (30% overhead)                 | Min: $26 \div 1.5 \times 1.30 \approx 22$ w. Max: $34 \div 1.5 \times 1.30 \approx 30$ w        |              |                     |
| Cost                                         | (avg salary \$250/week) $\times$ 1.5 devs $\times$ 30 weeks = <b>\$11,250</b>                   |              |                     |
| Profit Margin (min 10%, max 30%)             | Min: $\$11,250 \times 1.10 = \mathbf{\$12,375}$ Max: $\$11,250 \times 1.30 = \mathbf{\$14,625}$ |              |                     |

**Note:** pw = person-week. The team has 5 developers. Although each user requirement may require 1–2 developers at a time, the remaining developers work concurrently on other independent requirements, testing, reviewing, documentation, and integration tasks. This parallel execution across all 5 team members optimizes development time. Average developer salary is estimated at \$250/week. The maximum schedule time is estimated at 30 weeks including 30% overhead. Final project price to the client is **\$14,625**.

## 2.4 Actors Analysis

**MODIFIED**

**Lead:** Wafaa Shaheen

**Contributors:** [Ellen Izz - reviewing], [Yousef Qawwas – discussion], [Noor Nasir – finalizing]

| Actor         | Description                                                                                                                                                                                                                                                                                                                    |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Guest</b>  | An unregistered visitor who can browse only public, non-personalised content such as service information, the booking guide, and the FAQ. A Guest cannot book appointments, make payments, or access any account-specific feature, and must create an account to become a Client.                                              |
| <b>Client</b> | A registered and authenticated user (pet owner or farmer) who requests veterinary services. A Client can log in, update account information, book, cancel, and reschedule visits, make payments, track visit status, message the assigned veterinarian, request online consultations, and submit reviews for completed visits. |

| Actor                       | Description                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Veterinarian</b>         | A registered professional whose account is activated only after Administrator approval. A Veterinarian sets availability, views and updates assigned appointments, acknowledges emergency bookings, submits post-visit medical records and prescriptions, messages clients, and may view medical history only for animals linked to their own appointments.                       |
| <b>Administrator</b>        | The pre-configured system owner who governs the platform: approves or rejects veterinarian registrations, updates user account status, configures service pricing and policies, moderates reviews, and generates performance reports. The Administrator account cannot be registered publicly.                                                                                    |
| <b>Payment Gateway</b>      | An external system that the Vet Doctor system invokes to authorise and process card payments and that returns an approval or rejection result. It is acted upon by the system, never initiates a use case, and holds no client data beyond the transaction itself.                                                                                                                |
| <b>Notification Service</b> | An external messaging system that the Vet Doctor system invokes to deliver outbound notifications such as booking confirmations, status updates, reminders, password-reset links, and invoices to Clients and Veterinarians. Like the Payment Gateway it is a supporting actor that the system communicates with to complete a use case; it never triggers a use case on its own. |

## 2.5 Use-Case Diagram

MODIFIED

Lead: Fadi Bassous

Contributors: [Wafaa Shaheen – reviewing], [Ellen Izz – discussion], [Yousef Qawwas - diagram], [Noor Nasir – finalizing]

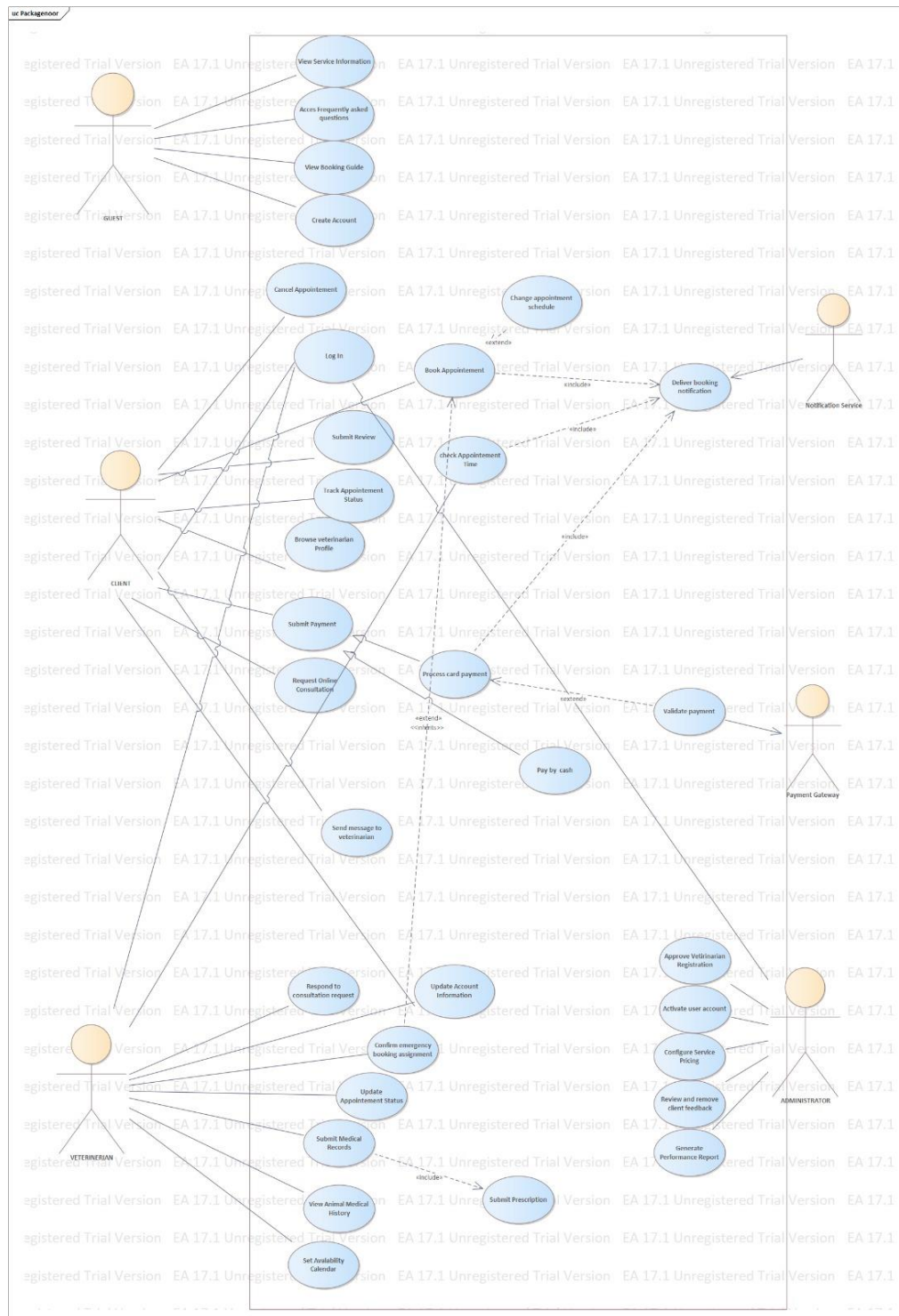


Figure 1 Overall use-case diagram of the Vet Doctor system .

## 2.6 Description of Key Use-Cases

MODIFIED

### Use Case 1: Book Appointment      Individual Task - Ellen Izz

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>System Name</b>                         | Vet Doctor System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Use Case Title</b>                      | Book Appointment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Description</b>                         | A Client requests a veterinary home or farm visit by specifying the service type, animal details, location, and preferred date and time. The system checks availability, assigns a veterinarian, creates the appointment, and notifies both parties.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Actors</b>                              | Client (primary); Veterinarian; Notification Service (secondary).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Data</b>                                | Service type, animal type, reason for visit, visit location, preferred date and time, veterinarian preference.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Stimulus/Trigger</b>                    | The Client selects "Book a Visit" on the dashboard to request a veterinary visit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Pre-conditions</b>                      | <ol style="list-style-type: none"><li>1. The Client is authenticated and has an active session.</li><li>2. At least one veterinarian whose specialisation and service area match the request has open availability.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Workflow OR Sequence/Flow of Events</b> | <ol style="list-style-type: none"><li>1. The Client selects a service type (Routine Check-Up, Emergency Visit, or Farm Visit).</li><li>2. The Client enters the animal details and the reason for the visit.</li><li>3. The Client enters the visit location and a preferred date and time.</li><li>4. The system checks availability for the requested slot.</li><li>5. The system confirms the slot is available.</li><li>6. The Client selects a specific veterinarian or requests automatic assignment.</li><li>7. The system displays the booking summary.</li><li>8. The Client confirms the booking.</li><li>9. The system creates the appointment record with status "Booked".</li><li>10. The system performs Send Notification to inform the Client and the assigned Veterinarian. [«include» Send Notification]</li></ol> |
| <b>Alternative Flows</b>                   | <p>A1 (at step 5) - Preferred slot unavailable: IF the requested slot is not available, the system offers the next three available slots; the Client selects one and the flow resumes at step 6.</p> <p>A2 (at step 6) - Automatic assignment: IF the Client does not choose a veterinarian, the system assigns the nearest available qualified veterinarian and the flow resumes at step 7.</p> <p>A3 (after step 9) - Emergency Visit: IF the service type is Emergency Visit, the system flags the booking as high priority and initiates Acknowledge Emergency Booking (15-minute deadline).</p>                                                                                                                                                                                                                                 |
| <b>Exception Flows</b>                     | <p>E1 (at step 4) - No availability: IF no veterinarian is available for the requested service type and date, the system informs the Client and the use case ends without creating a booking.</p> <p>E2 (at step 8) - Session expired: IF the Client session has expired before confirmation, the system discards the draft, prompts re-authentication, and the use case ends without creating a booking.</p>                                                                                                                                                                                                                                                                                                                                                                                                                        |



|                                |                                                                                                                                                                                                                                                                                                              |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Postconditions/Response</b> | <ol style="list-style-type: none"> <li>1. A new appointment record exists with status "Booked".</li> <li>2. The Client and the assigned Veterinarian have each received a booking notification.</li> <li>3. On any exception path no appointment record is created and system state is unchanged.</li> </ol> |
| <b>Comments</b>                | Emergency Visit bookings are high priority. A cancellation made less than two hours before the visit triggers a cancellation-fee notification (see Cancel Appointment).                                                                                                                                      |

## Use Case 2: Log In      Individual Task - Noor Nasir

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>System Name</b>                         | Vet Doctor System                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Use Case Title</b>                      | Log In                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Description</b>                         | A registered Client, Veterinarian, or Administrator authenticates with a registered email and password and is granted access to a role-specific interface.                                                                                                                                                                                                                                                                                                       |
| <b>Actors</b>                              | Client, Veterinarian, Administrator (primary); Notification Service (secondary).                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Data</b>                                | Registered email address, password.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Stimulus/Trigger</b>                    | A registered user submits their credentials on the login screen.                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Pre-conditions</b>                      | <ol style="list-style-type: none"> <li>1. The user has a registered account in the system.</li> </ol>                                                                                                                                                                                                                                                                                                                                                            |
| <b>Workflow OR Sequence/Flow of Events</b> | <ol style="list-style-type: none"> <li>1. The user enters their registered email and password.</li> <li>2. The user submits the login request.</li> <li>3. The system validates the credentials against the stored account.</li> <li>4. The system confirms the credentials are valid and the account is active.</li> <li>5. The system creates an authenticated session.</li> <li>6. The system redirects the user to their role-specific interface.</li> </ol> |
| <b>Alternative Flows</b>                   | A1 (at step 1) - Forgot password: IF the user selects "Forgot Password", the system sends a reset link valid for 30 minutes via the Notification Service, and the use case ends after the link is sent.                                                                                                                                                                                                                                                          |
| <b>Exception Flows</b>                     | <p>E1 (at step 3) - Invalid credentials: IF the email or password is incorrect, the system shows an error and returns to step 1; after 5 consecutive failures it locks the account for 15 minutes and sends a lockout email via the Notification Service.</p> <p>E2 (at step 4) - Inactive account: IF the account is locked, suspended, or pending Administrator approval, the system rejects the login, states the reason, and the use case ends.</p>          |
| <b>Postconditions/Response</b>             | <ol style="list-style-type: none"> <li>1. The user is authenticated and an active session has been created.</li> <li>2. The user is viewing their role-specific interface.</li> <li>3. On any exception path no session is created and the user remains unauthenticated.</li> </ol>                                                                                                                                                                              |

|                 |                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Comments</b> | Veterinarian accounts pending approval cannot log in. Passwords are stored using a non-reversible hash and are never transmitted in plain text. |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

### Use Case 3: Make Payment      Individual Task - Yousef Qawwas

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>System Name</b>                         | Vet Doctor System                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Use Case Title</b>                      | Make Payment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Description</b>                         | A Client pays for a veterinary service by card or cash. The system calculates the invoice, processes the payment, generates a digital invoice, and updates the appointment record.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Actors</b>                              | Client (primary); Payment Gateway, Notification Service (secondary); Veterinarian (confirms cash receipt).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Data</b>                                | Appointment ID, invoice ID, payment method, base service charge, post-visit charges, total invoice amount, invoice status, appointment payment status, transaction reference, payment timestamp, receipt confirmation, and masked card reference.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Stimulus/Trigger</b>                    | The Client selects Make Payment for a completed appointment that has an unpaid final invoice.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Pre-conditions</b>                      | <ol style="list-style-type: none"> <li>1. The Client is authenticated with an active session.</li> <li>2. The appointment belongs to the logged-in Client.</li> <li>3. The appointment status is <b>Completed</b>.</li> <li>4. The final service charges have been recorded.</li> <li>5. The invoice status is <b>Unpaid</b>.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Workflow OR Sequence/Flow of Events</b> | <ol style="list-style-type: none"> <li>1. The Client selects Make Payment for the completed appointment.</li> <li>2. The system verifies that the appointment is completed and unpaid.</li> <li>3. The system retrieves or creates the final invoice and calculates the total amount from the base price and post-visit charges.</li> <li>4. The system displays the itemized invoice to the Client.</li> <li>5. If additional charges exist, the system asks the Client to acknowledge them before continuing.</li> <li>6. The Client confirms the invoice details.</li> <li>7. The system displays the available payment methods.</li> <li>8. The Client selects Pay by Credit/Debit Card.</li> <li>9. The system creates a payment request and forwards it to the Payment Gateway.</li> <li>10. The Payment Gateway displays a secure payment form.</li> <li>11. The Client enters card details through the Payment Gateway.</li> <li>12. The Payment Gateway validates and processes the payment.</li> <li>13. If approved, the gateway returns the approval result, transaction reference, and masked card reference.</li> <li>14. The system records the payment, updates the invoice status to Paid, and updates the appointment payment status to Paid.</li> <li>15. The system generates and stores the digital invoice/receipt.</li> </ol> |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | <p>16. The system requests the Notification Service to send the payment confirmation and invoice link.</p> <p>17. The system displays a successful payment confirmation to the Client.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Alternative Flows</b>       | <p>A1 - Pay by Cash: At step 8, if the Client selects Pay by Cash, the system marks the invoice as Cash Pending and displays the amount due. The Client pays the Veterinarian in cash. After the Veterinarian confirms receipt, the system records the cash payment, updates the invoice and appointment payment statuses to Paid, generates the receipt, and sends the confirmation notification.</p> <p>A2 - Additional Charges Require Acknowledgement: At step 5, if additional charges exist, the system displays their reason and amount. If the Client acknowledges them, the use case continues. If not, the system cancels the payment process and keeps the invoice Unpaid.</p> |
| <b>Exception Flows</b>         | <p>E1 - Card Payment Rejected: If the Payment Gateway rejects the payment, the system records the failed attempt, keeps the invoice Unpaid, and allows the Client to retry or choose another method.</p> <p>E2 - Gateway Unavailable or Timeout: If the gateway is unavailable or does not respond, the system records the failure, keeps the invoice Unpaid, and displays a temporary failure message.</p> <p>E3 - Cash Receipt Not Confirmed: If the Veterinarian does not confirm cash receipt, the system keeps the invoice Cash Pending and does not generate a paid receipt.</p>                                                                                                    |
| <b>Postconditions/Response</b> | <p>1. If payment succeeds, the payment is recorded and linked to the invoice and appointment.</p> <p>2. If payment succeeds, the invoice and appointment payment statuses are updated to Paid.</p> <p>3. If payment succeeds, a digital invoice/receipt is generated, stored, and sent to the Client.</p> <p>4. If payment fails, is cancelled, or cash receipt is not confirmed, no successful payment record or paid receipt is created, and the invoice remains Unpaid or Cash Pending.</p>                                                                                                                                                                                            |
| <b>Comments</b>                | <p>Full card numbers are never stored by the Vet Doctor system. Card details are entered through the Payment Gateway, and the system only stores the transaction reference and masked card reference. Clients can filter and download past invoices as PDF files.</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |

#### Use Case 4: Acknowledge Emergency Booking      Individual Task - Wafaa Shaheen

|                       |                                                                                                                                                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>System Name</b>    | Vet Doctor System                                                                                                                                                                                               |
| <b>Use Case Title</b> | Acknowledge Emergency Booking                                                                                                                                                                                   |
| <b>Description</b>    | When an Emergency Visit booking is created, the system notifies the assigned Veterinarian, who must acknowledge within 15 minutes. If acknowledgement does not occur in time, the system reassigns the booking. |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>                              | Veterinarian (primary); Client, Notification Service, administrator (secondary).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Data</b>                                | Appointment ID, booking type (Emergency Visit), client location, animal details, reason for visit, acknowledgement timestamp, assignment status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Stimulus/Trigger</b>                    | The system flags a newly created Emergency Visit booking as high priority and notifies the assigned Veterinarian.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Pre-conditions</b>                      | <ol style="list-style-type: none"> <li>1. An Emergency Visit booking exists and has been flagged as high priority.</li> <li>2. A qualified Veterinarian has been assigned to the booking.</li> <li>3. The assigned Veterinarian has an active account.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Workflow OR Sequence/Flow of Events</b> | <ol style="list-style-type: none"> <li>1. The system sends an emergency-booking notification to the assigned Veterinarian via the Notification Service, including the client location, animal details, reason for visit, and the 15-minute deadline.</li> <li>2. The Veterinarian reviews the emergency-booking details.</li> <li>3. The Veterinarian acknowledges the booking within 15 minutes.</li> <li>4. The system records the acknowledgement timestamp and sets the appointment status to "Veterinarian Assigned".</li> <li>5. The system performs Send Notification to confirm to the Client that a veterinarian has accepted. [«include» Send Notification]</li> </ol> |
| <b>Alternative Flows</b>                   | A1 (at step 3) - Veterinarian declines: IF the assigned Veterinarian explicitly declines, the system proceeds immediately to reassignment (see E1) without waiting for the deadline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Exception Flows</b>                     | <p>E1 (at step 3) - Acknowledgement timeout: IF the Veterinarian does not acknowledge within 15 minutes, the system reassigns the booking to the next available qualified Veterinarian, notifies the Client and the original Veterinarian, and restarts the deadline.</p> <p>E2 (at step 3 / E1) - No alternative veterinarian: IF no other qualified Veterinarian is available, the system notifies the Client that none is currently available and escalates the booking to the Administrator.</p>                                                                                                                                                                             |
| <b>Postconditions/Response</b>             | <ol style="list-style-type: none"> <li>1. The booking has been acknowledged with status "Veterinarian Assigned", or it has been reassigned or escalated.</li> <li>2. The outcome is timestamped and the Client has been notified.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Comments</b>                            | The 15-minute window is strictly enforced to guarantee a timely emergency response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

### Use Case 5: Submit Review      Individual Task - Fadi Bassous

|                       |                                                                                                                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>System Name</b>    | Vet Doctor System                                                                                                                                                                      |
| <b>Use Case Title</b> | Submit Review                                                                                                                                                                          |
| <b>Description</b>    | A Client submits a star rating and an optional written review for a completed visit. The system moderates the content, stores the review, and updates the veterinarian average rating. |
| <b>Actors</b>         | Client (primary); Notification Service (secondary).                                                                                                                                    |
| <b>Data</b>           | Star rating (1 to 5), optional review text (up to 500 characters), appointment                                                                                                         |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | reference, submission timestamp.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Stimulus/Trigger</b>                    | The Client selects "Leave a Review" for a completed appointment, or follows the review prompt sent after the visit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Pre-conditions</b>                      | <ol style="list-style-type: none"> <li>1. The Client is authenticated with an active session.</li> <li>2. The referenced appointment is marked as Completed.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Workflow OR Sequence/Flow of Events</b> | <ol style="list-style-type: none"> <li>1. The system presents the review form for the completed appointment.</li> <li>2. The Client enters a star rating from 1 to 5.</li> <li>3. The Client optionally enters a written review (up to 500 characters).</li> <li>4. The Client submits the review.</li> <li>5. The system runs an automated content-moderation check.</li> <li>6. The system stores the review and links it to the appointment and the Veterinarian.</li> <li>7. The system recalculates the Veterinarian average rating.</li> <li>8. The system displays a confirmation to the Client.</li> </ol> |
| <b>Alternative Flows</b>                   | A1 (at step 5) - Flagged content: IF the moderation check flags inappropriate language, the system withholds the review from publication and queues it for Administrator moderation (see Moderate Review); the Client is told the review is pending.                                                                                                                                                                                                                                                                                                                                                               |
| <b>Exception Flows</b>                     | <p>E1 (at step 1) - Duplicate review: IF the Client has already reviewed this appointment, the system blocks a second submission and the use case ends.</p> <p>E2 (at step 2) - Invalid rating: IF no rating between 1 and 5 is provided, the system rejects the submission and prompts the Client to supply a valid rating.</p>                                                                                                                                                                                                                                                                                   |
| <b>Postconditions/Response</b>             | <ol style="list-style-type: none"> <li>1. Exactly one review is recorded for the appointment and linked to the Veterinarian.</li> <li>2. The Veterinarian average rating has been recalculated for a published review, or the review is held pending moderation.</li> </ol>                                                                                                                                                                                                                                                                                                                                        |
| <b>Comments</b>                            | Only registered Clients may review, and only completed appointments are eligible. Each appointment may receive only one review.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## 2.7 Activity Diagram

MODIFIED

Lead: Yousef Qawwas

Contributors: [Ellen Izz – reviewing], [Fadi Bassous – discussion], [Wafaa Shaheen – finalizing], [Noor Nasir – diagram]

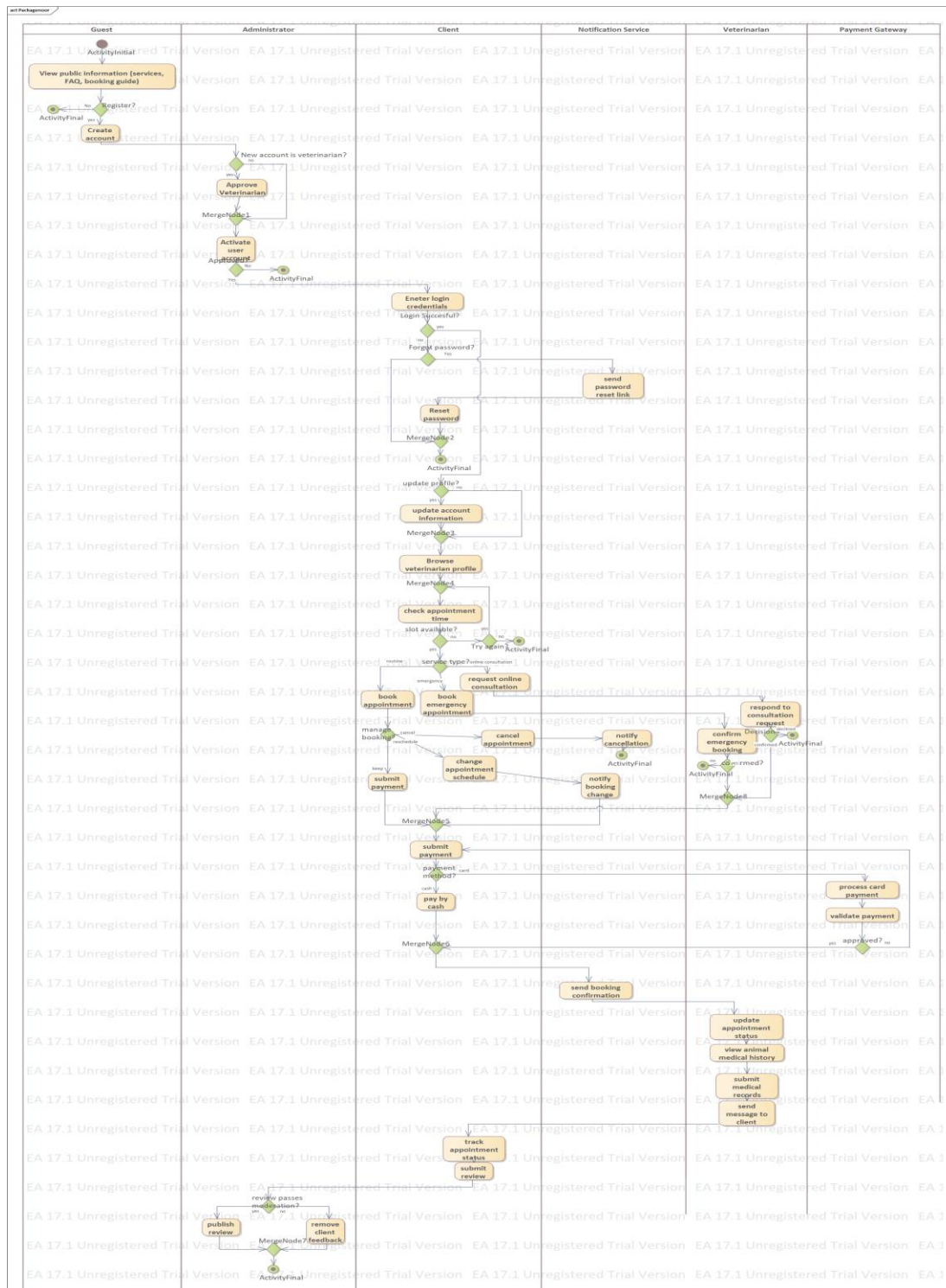


Figure 2 overall activity diagram.

## 3. System Analysis and Modelling

MODIFIED

### 3.1 System Class Diagrams

Lead: Fadi Bassous

Contributors: [Yousef Qawwas – reviewing], [Noor Nasir – discussion], [Ellen Izz – finalizing], [Wafaa Shaheen – diagram]

#### 3.1.1 Analysis Class Model

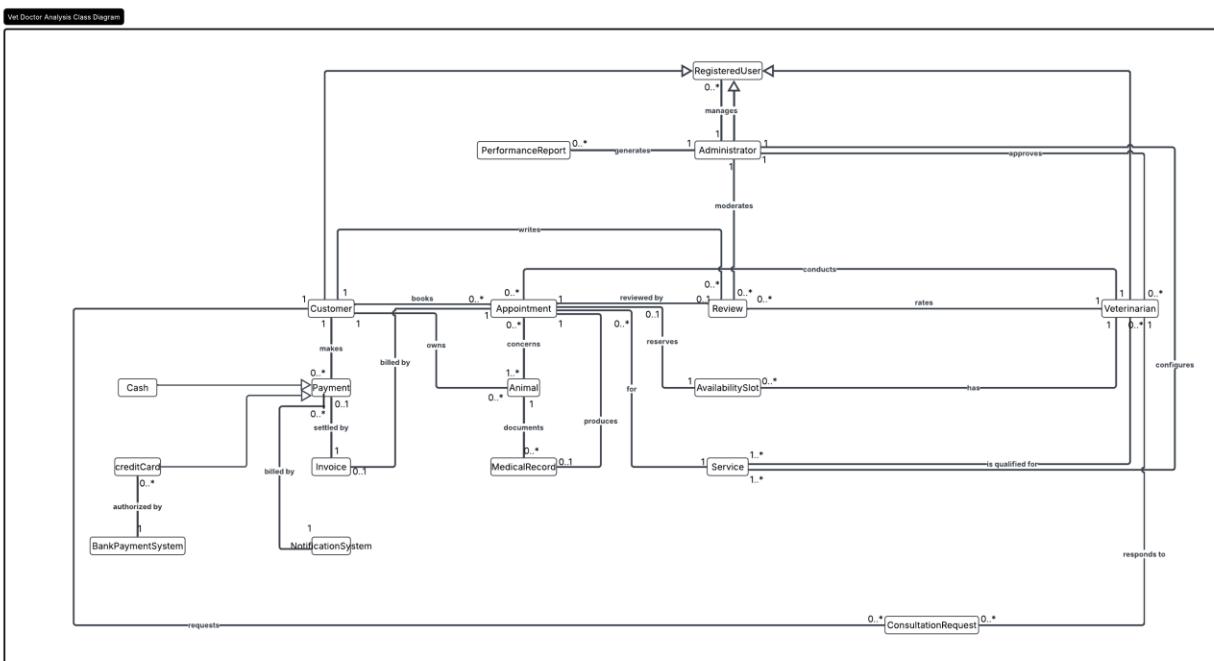


Figure 3: Analysis class model.

#### 3.1.2 Detailed Class Model

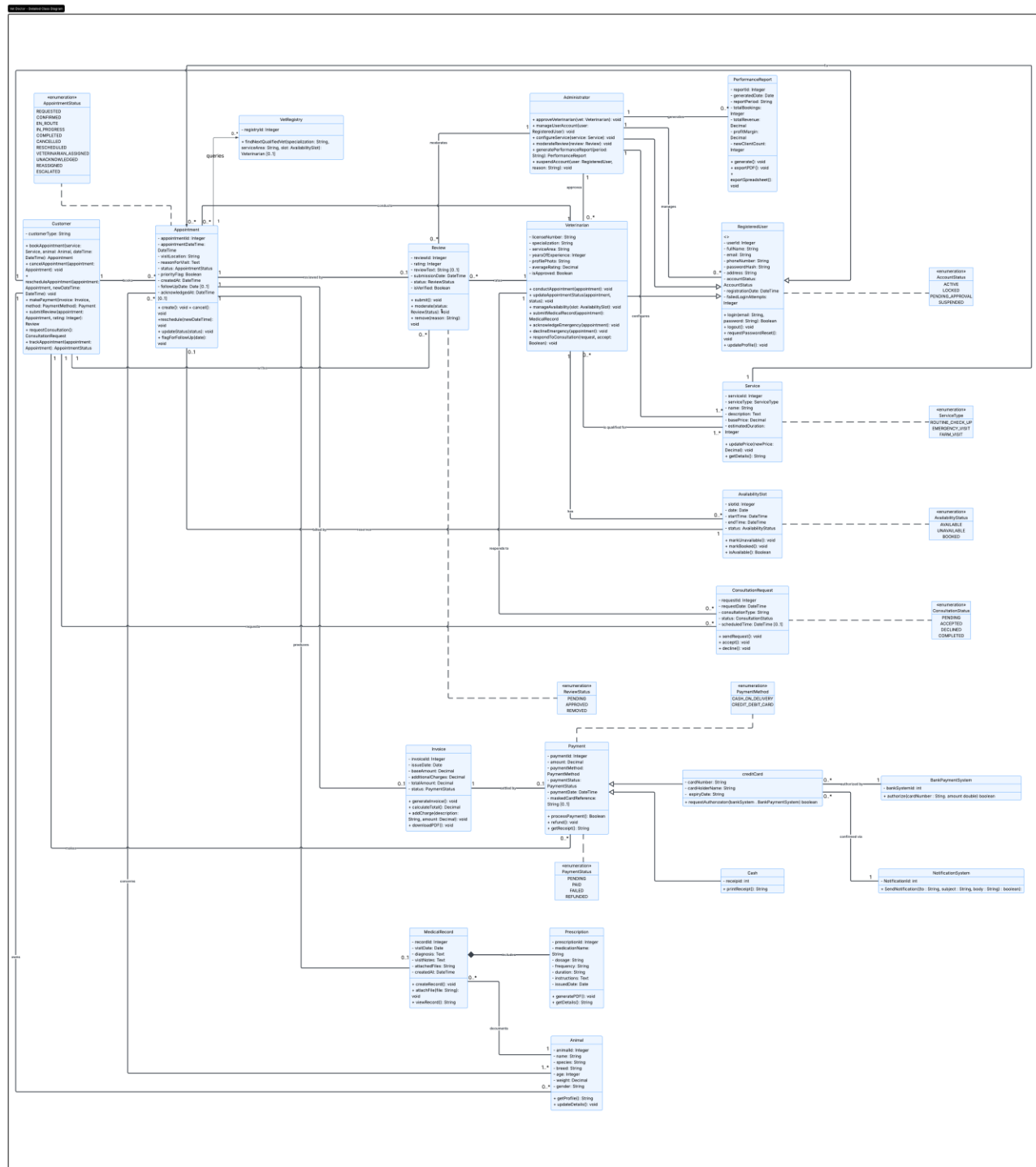


Figure 4: Detailed class model.



## 3.2 Sequence Diagrams

### 1. Use Case #1: Book Appointment [Ellen Izz]

1.

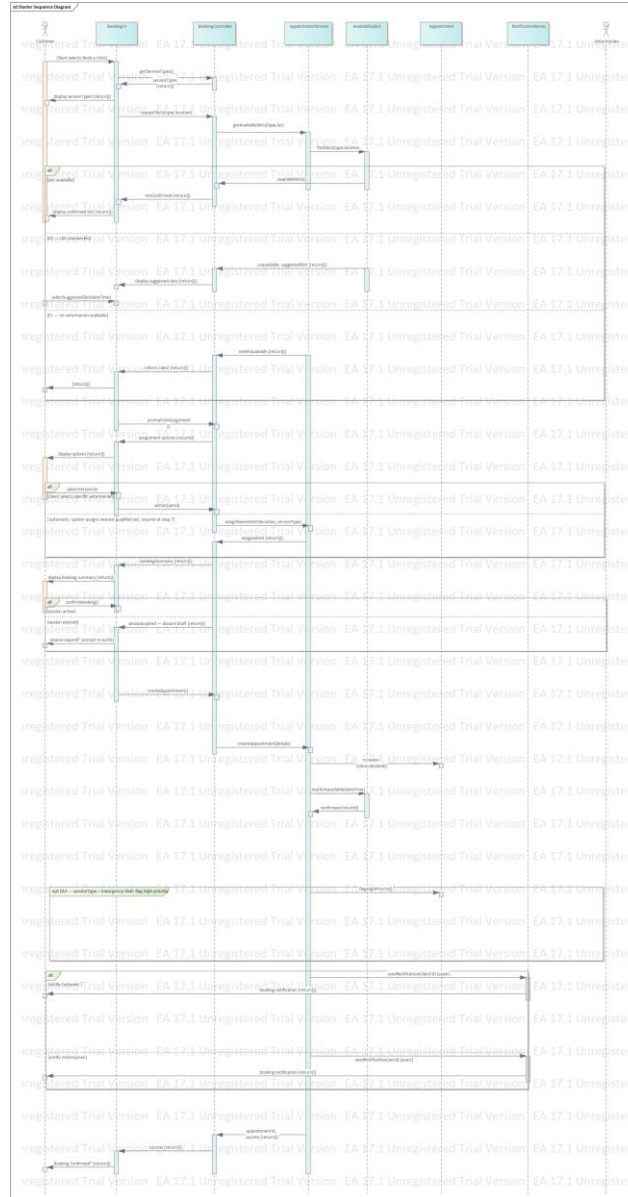


Figure 5: Book Appointment Use Case

## 2. Use Case #2: Login [Noor Nasir]

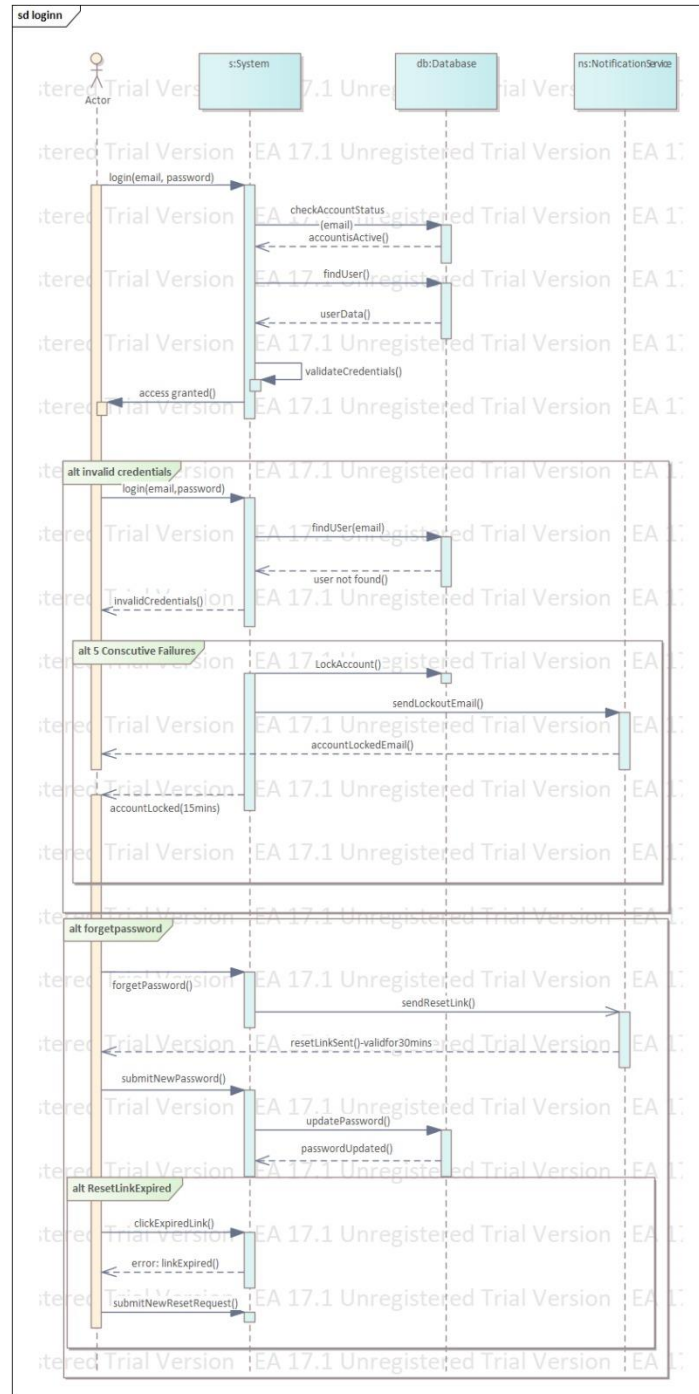


Figure 6: Login Use Case

### 3. Use Case #3: Make Payment [Yousef Qawwas]

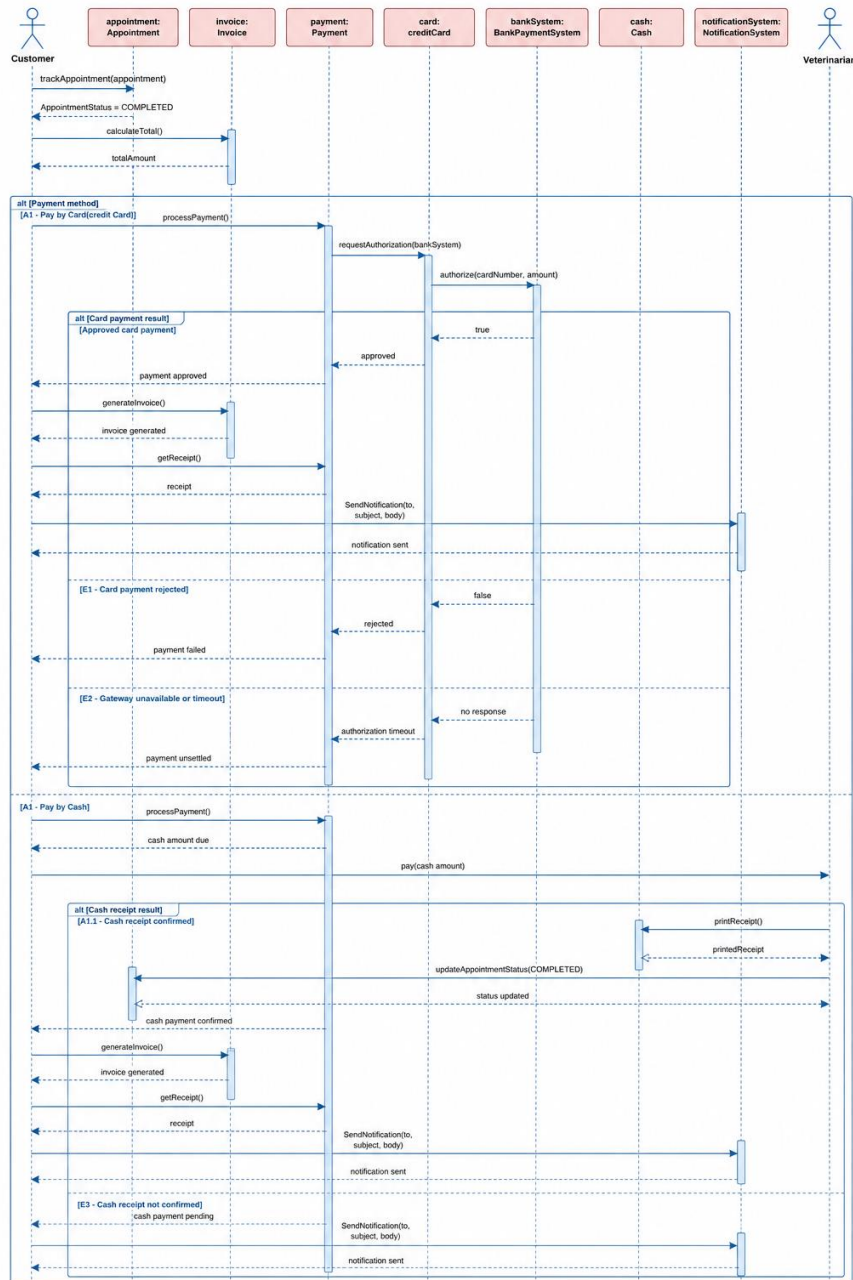


Figure 7: Make Payment Use Case

#### 4. Use Case #4: Acknowledge Emergency Booking [Wafaa Shaheen]

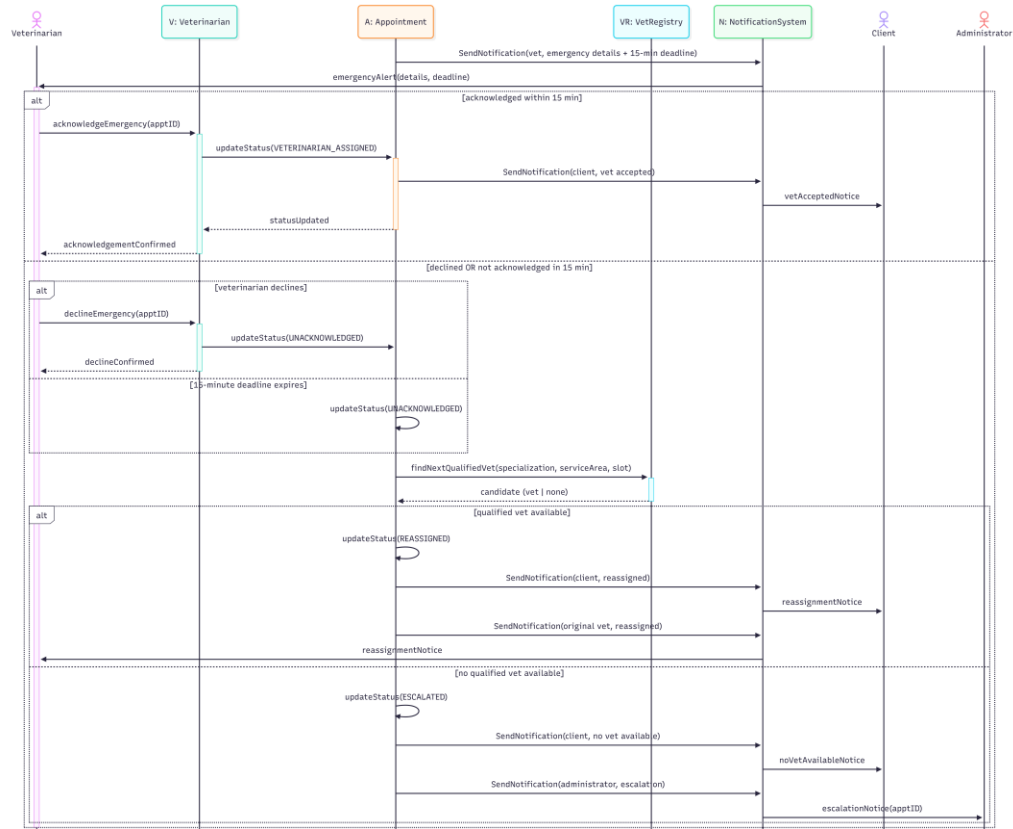


Figure 8: Acknowledge Emergency Booking Use Case

## 5. Use Case #5: Submit Review [Fadi Bassous]

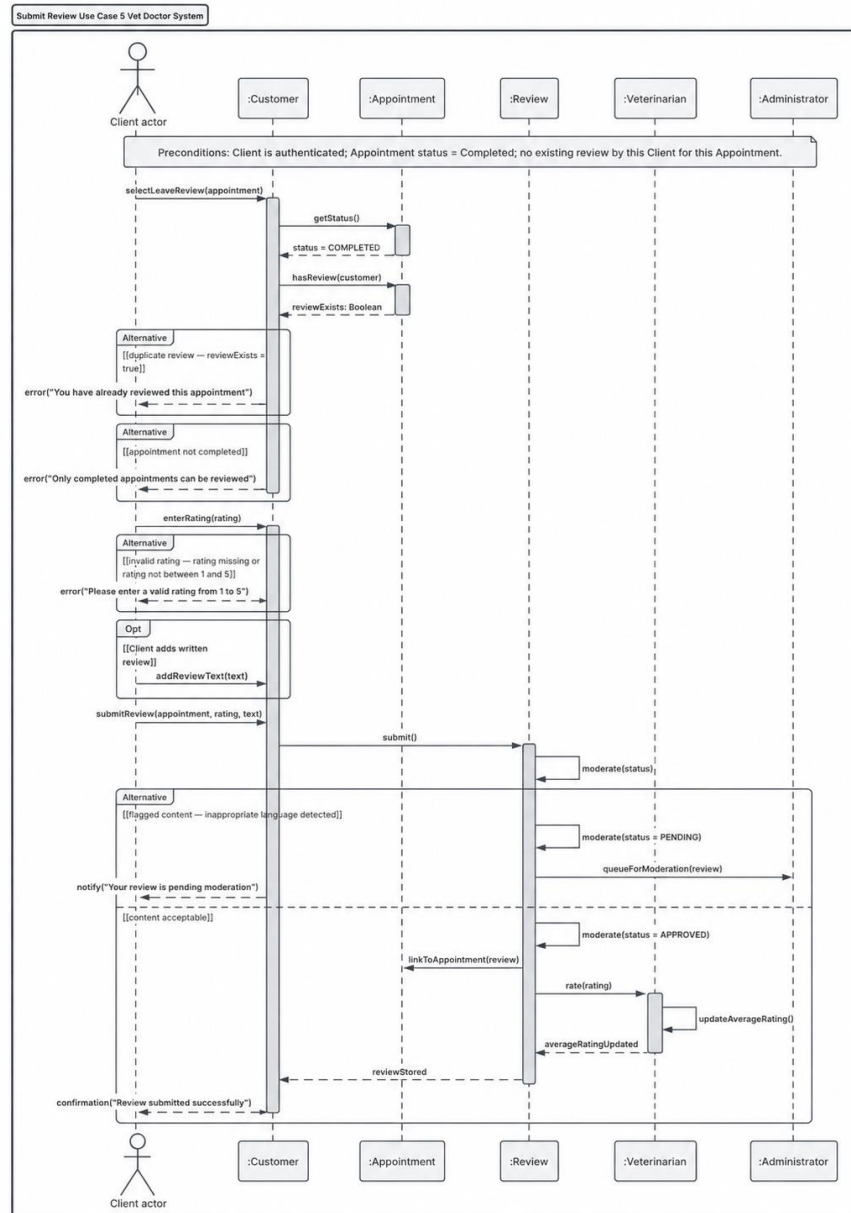


Figure 9: Acknowledge Emergency Booking

## 4. System Design and Modelling

### 4.1 Description of chosen Design Goals

**MODIFIED**

Lead: Ellen Izz

Contributors: [Fadi Bassous – reviewing], [Wafaa Shaheen – discussion], [Yousef Qawwas – finalizing]

The following design goals guide the architecture of the Vet Doctor system. Two are general engineering principles that keep the system maintainable, and a specific goal derived directly from the system's non-functional requirements and stated in measurable terms so it can be tested and verified.

#### 6. Low Coupling (general goal)

The components of the system should depend on one another as little as possible, so that a change in one component does not force changes in the others. Keeping dependency low means each module can be developed, tested, replaced, and maintained on its own.

Example: if the payment-processing module is modified for instance, to add a new e-wallet provider the appointment-booking module must keep working without any change, because the two communicate only through a well-defined interface rather than sharing internal data.

#### 7. High Cohesion (general goal)

Each component should contain only the responsibilities that serve a single, clearly defined purpose, so that everything inside a module contributes to one job.

Example: the veterinarian-scheduling module should handle only availability calendars, slot management, and appointment status. It should not also calculate invoices or moderate reviews, since those responsibilities belong to the payment module and the feedback module respectively.

#### 8. Performance (specific goal)

The system must respond fast enough to meet the time-bound needs of the business, especially emergencies. An Emergency Visit booking must be flagged as high priority and the assigned veterinarian notified immediately, so that the required 15-minute acknowledgement window is met. A booking confirmation must be generated and dispatched to both the client and the veterinarian within 30 seconds of a successful booking, and any change to an appointment's status En Route, In Progress, or Completed must be reflected to the client in real time. Because each of these is stated as a concrete time threshold, the goal is verifiable: response and notification times can be measured directly against them.

## 4.2 Component Diagram of your system components

Lead: Ellen Izz

Contributors: [Wafaa Shaheen – reviewing], [Fadi Bassous – discussion], [Yousef Qawwas – finalizing], [Ellen Izz – diagram]

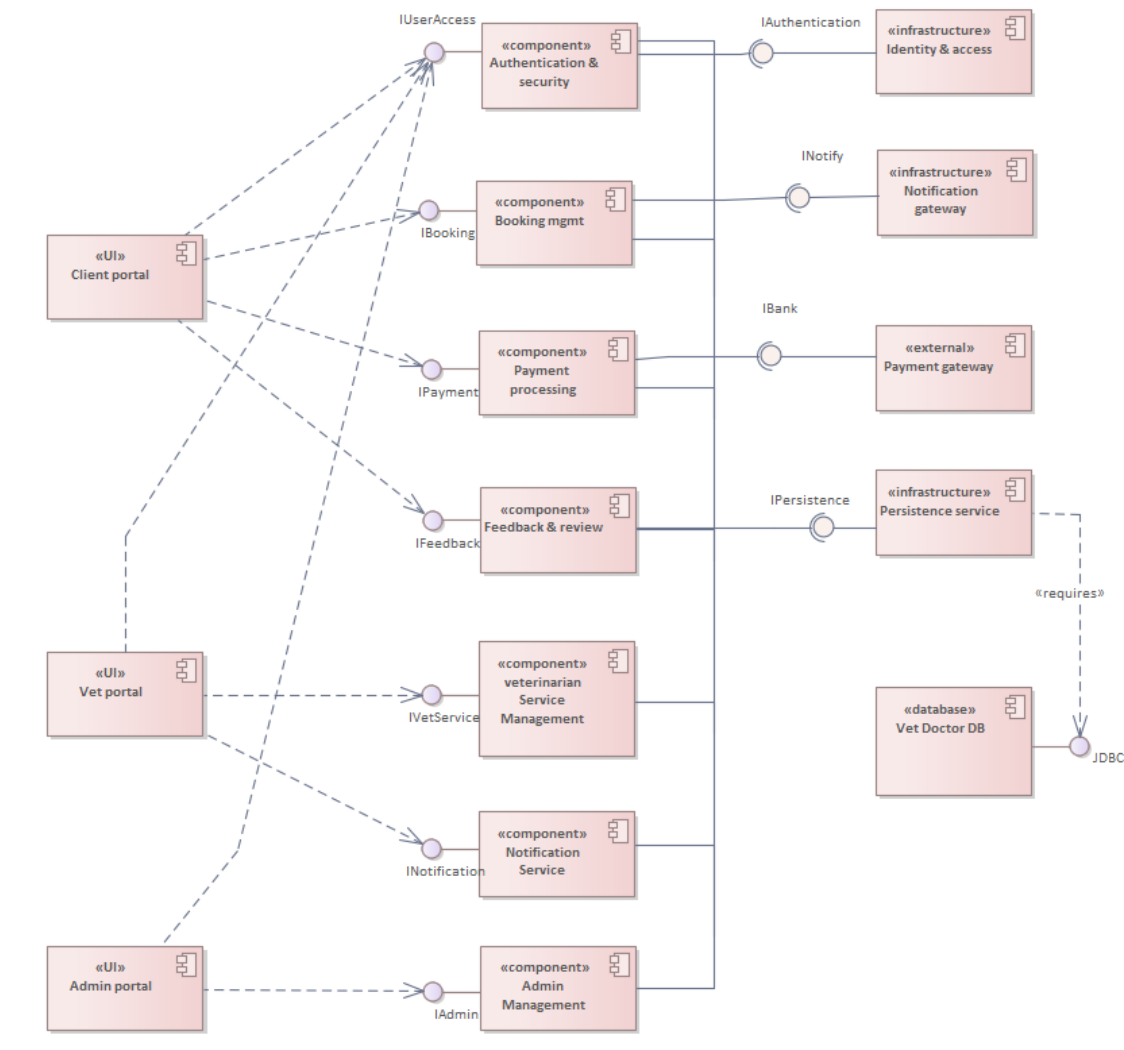


Figure 10: Component Diagram

The component diagram shows the main software components of the Vet Doctor system and how they interact through defined interfaces. The Client, Vet, and Admin portals connect to the middle-layer components according to their roles, such as authentication, booking, payment, feedback, notification, vet service, and admin management. These components then communicate with infrastructure services, including identity access, payment gateway, notification gateway, and persistence service. The persistence service connects to the Vet Doctor database through JDBC. This design helps organize system

responsibilities, reduce direct dependencies between components, and make the system easier to maintain and extend.

### 4.3. Overall architecture diagram

Lead: Yousef Qawwas

Contributors: [Noor Nasir– reviewing], [Ellen Izz – discussion], [Fadi Bassous - finalizing]

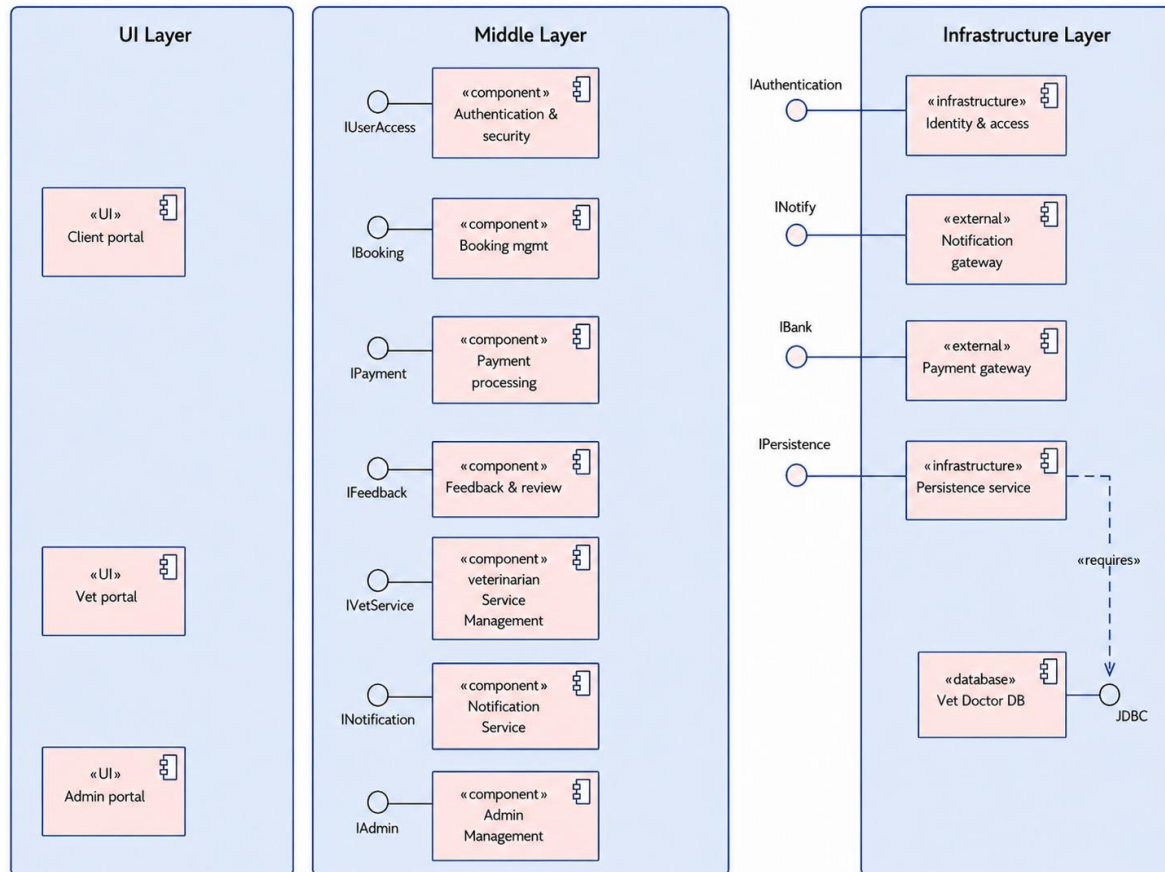


Figure 11: Overall architecture diagram

The Vet Doctor system follows a layered client–server architecture. This architecture was chosen because it separates the system into clear parts and supports the main design goals: low coupling, high cohesion, and performance. The system is divided into three main layers: the UI Layer, the Middle Layer, and the Infrastructure Layer.

The UI Layer contains the Client Portal, Vet Portal, and Admin Portal. This layer is responsible only for user interaction, such as displaying information, receiving requests, and sending them to the system. It does not directly handle business logic or database operations.



The Middle Layer contains the core business components of the system, including Authentication and Security, Booking Management, Payment Processing, Feedback and Review, Veterinarian Service Management, Notification Service, and Admin Management. Each component handles a specific function in the system and processes requests coming from the UI layer.

The Infrastructure Layer provides the supporting technical services needed by the system. It includes Identity and Access, the external Payment Gateway, the external Notification Gateway, the Persistence Service, and the Vet Doctor Database. These services support authentication, online payment, notifications, and permanent data storage.

### **Design Goal 1: Low Coupling**

Low coupling is achieved by separating the system into independent layers and components. The UI portals do not directly access the database or external gateways. Instead, they communicate with the middle-layer components. Also, components such as Payment Processing, Notification Service, and Persistence Service communicate through defined interfaces. For example, if the payment provider is changed or an e-wallet provider is added, the Booking Management component should not be affected because payment operations are handled separately.

### **Design Goal 2: High Cohesion**

High cohesion is achieved by giving each component one clear responsibility. Booking Management handles appointment booking, Payment Processing handles invoices and payments, Feedback and Review handles customer feedback, and Veterinarian Service Management handles veterinarian availability, service details, and appointment status. This makes the system easier to understand, test, maintain, and extend because each component focuses on one main task.

### **Design Goal 3: Performance**

Performance is supported by separating time-sensitive operations into clear components. Emergency Visit bookings can be handled by Booking Management and flagged as a high priority, while Notification Service can immediately notify the assigned veterinarian. Booking confirmations can also be sent quickly through the Notification Gateway. In addition, appointment status updates such as En Route, In Progress, and Completed can be processed by the related service components and reflected to the client without affecting unrelated parts of the system.

Overall, the layered client–server architecture gives the Vet Doctor system a clear structure. It separates user interfaces, business logic, external services, and database storage, which helps the system remain maintainable, organized, and able to meet its main performance requirements.

## 4.4. Deployment diagram

**MODIFIED**

Lead: Noor Nasir

Contributors: [Wafaa Shaheen– reviewing], [Ellen Izz– discussion], [Yousef Qawwas – finalizing]

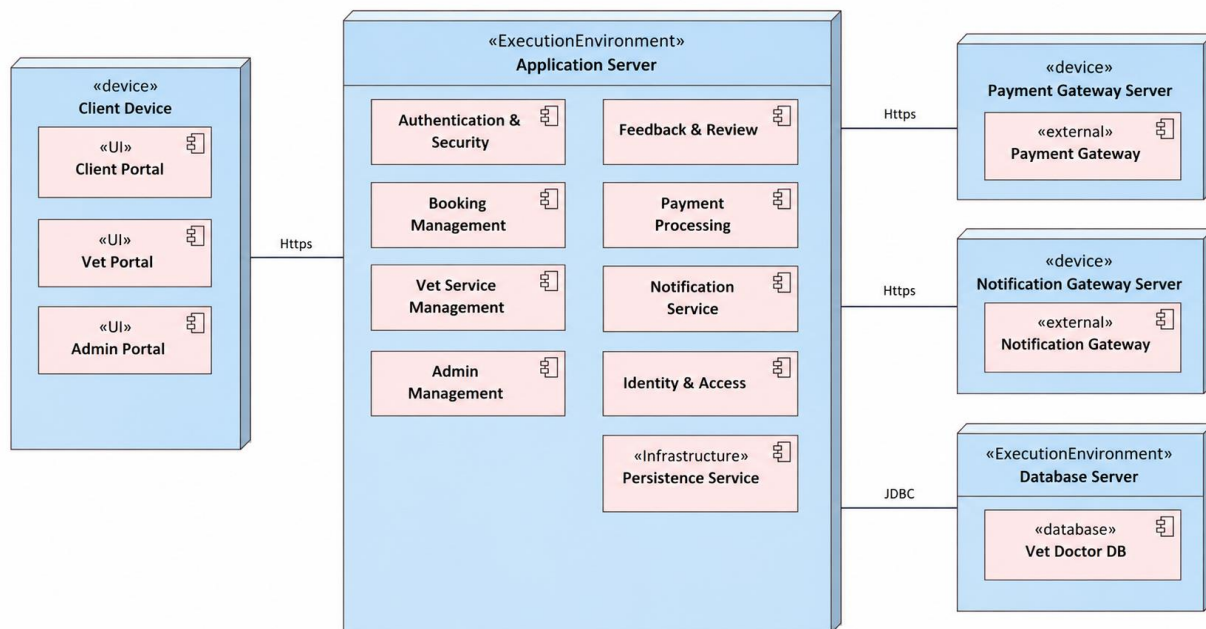


Figure 12: Deployment diagram

The deployment diagram shows the implementation structure of the Vet Doctor system. The Client Device contains the Client, Vet, and Admin portals, which connect to the Application Server through HTTPS. The Application Server hosts the main business components, while the Payment Gateway and Notification Gateway are deployed separately as external services. The Database Server stores the Vet Doctor DB and is accessed through the Persistence Service using JDBC. This structure separates user interfaces, business logic, external services, and data storage for better performance and maintainability.

## Appendix-I: Minutes of meetings

| Meeting No. | Date & Time            | Location | Attendees              | Topics Discussed                                                                                                                                                                                                                          | Outcomes                                                                                                                                                                              |
|-------------|------------------------|----------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1           | 25/04/2026,<br>6:00 PM | Online   | Members of both groups | Discussed business overview, service types (Routine Check-Up, Emergency Visit, Farm Visit), user types (pet owners, farmers, veterinarians), and booking process                                                                          | Confirmed three service types and main system actors. Agreed on business scope and services to be supported by the system                                                             |
| 2           | 26/4/2026<br>8:00 PM   | Online   | Our team members       | Discussed the received business description, identified the main business services, reviewed the expected tasks, and divided the work among team members.                                                                                 | Agreed on the initial understanding of the Vet Doctor business, identifying the main actors as clients, veterinarians, and administrators.                                            |
| 3           | 28/4/2026<br>6:30 PM   | Online   | Our team members       | Reviewed the first draft of the user requirements, checked whether each requirement represented one business service, discussed possible system requirement details, and prepared questions for the next meeting with the customer group. | Updated the wording of the initial user requirements, agreed on the main system requirement categories for each user requirement, and prepared discussion points about several tasks. |
| 4           | 3/5/2026,<br>7:00 PM   | Online   | Members of both groups | Reviewed drafted user and system requirements UR1-UR9, discussed emergency visit priority handling, payment methods, admin responsibilities, and veterinarian approval process                                                            | Agreed on final user requirements and system requirements. Confirmed payment options and emergency handling approach                                                                  |

In addition to the recorded meetings listed above, several informal meetings and discussions were held among team members, either online or during university breaks. Some short discussions were also held with the customer group between classes and after class time. These meetings were not recorded with exact dates and times, but they were used to clarify unclear points across the different project phases, discuss requirement details, review feedback, agree on updates to diagrams and specifications, and coordinate the distribution of work among team members. The outcomes of these informal discussions helped the team agree on several design and requirement decisions, resolve ambiguities, improve consistency between individual contributions, and prepare the final deliverables in a more organized form.